

BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer’s advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower BIRCH semantic store. Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy. Exact replay is reported only as a surface-form reference; the main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level index Soft O2 improves over dense FlatIP, with hard hydration and shuffled-*sid* controls separating binding from score copying and membership noise. Gains are largest where a partial episode hit can pull in its answer sibling. Soft O2-C and gated Hybrid extend binding to BIRCH clusters under a Mix protocol with off-session gold; when query and gold already share a session, Soft O2 is preferred. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a complete/incomplete/miss failure taxonomy, and paired significance tests.

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1. Introduction

A junior associate may ask a firm assistant: “Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?” The system returns several near-identical “contract termination” questions, ranks the stored user utterance highly, and never surfaces the partner’s written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology, neighbouring matters are topically adjacent yet legally distinct, and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal large language models (LLMs) further improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR), however, mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still need a reliable index of *internal consultation memory*: prior client-specific advice buried in a growing same-domain archive under lexical interference from neighbouring matters.

Under that interference, dense turn retrieval often returns a question turn while its answer sibling remains outside top- k . We term this failure *episode incompleteness*, as distinct from a full session miss. It is worsened when approximate indexes distort turn scores and fragment episode context at moderate archive sizes. Parent-document hydration and session-max aggregation can force siblings into the result list by hard score copy, but they also erase rank competition among unrelated candidates. What is needed is a way to complete episodes without collapsing the retrieval ranking into a session dump.

This paper presents BIMS-LEGAL, a dual-store system for Chinese legal consultation logs. A fast episodic pathway indexes timestamped turn embeddings with speaker role and session identifier *sid*; a slower semantic pathway

consolidates turns with Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) Zhang et al. (1996). The dual-pathway split is inspired by Complementary Learning Systems theory McClelland et al. (1995); Kumaran et al. (2016) only as an architectural analogy. On the episodic pathway, Soft O2 performs session soft binding: when a turn is retrieved with score s , siblings sharing sid inherit $\max(s_\nu, \beta \cdot s)$, so an answer turn can surface under question-hit / answer-miss while attenuated scores keep candidate competition intact. On the semantic pathway, Soft O2-C applies the same rule within BIRCH clusters when gold evidence is not co-session with the query, and a gated Hybrid triggers cluster binding only on direct dense hits. Exact FlatIP indexing (O1) restores faithful turn scores when product quantization collapses Answer Hit; bulk-load consolidation (O3) makes large shared-store construction tractable without changing retrieval semantics. We study how often episode incompleteness occurs relative to full session miss, how large Soft O2 gains are against FlatIP and hard hydration on paraphrase, follow-up, and advice-recall channels, and when Soft O2-C /Hybrid helps under Mix reconstructions with off-session gold.

The main contributions are as follows. First, we formulate consultation-memory retrieval under same-domain interference and identify episode incompleteness as a recurrent failure mode of dense turn retrieval. Second, we propose Soft O2 as a turn-level soft-binding operator, with a gap-ratio account of attenuation β and default $\beta=0.98$ chosen by validation sweep, supported by O1/O3 as implementation operators for shared-store evaluation. Third, we evaluate Soft O2 on CAIL multi-turn and rebuilt LegalEp corpora against FlatIP, hard hydration, shuffled- sid , BM25-turn, RRF, and cross-encoder controls, and examine Soft O2-C /Hybrid under Mix cross-session reconstructions with graded nDCG, a failure taxonomy, and paired significance tests.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL and the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-

71 Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware
 72 similar-case ranking further show how interaction context and structured le-
 73 gal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024).
 74 Those pipelines optimize access to external authority. Consultation memory
 75 retrieval targets a different evidence source: *prior consultation episodes* inside
 76 an agent’s own archive. Confusing two clients’ histories can produce fluent
 77 yet professionally inappropriate responses even when statute retrieval is cor-
 78 rect. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM
 79 improve statutory and advisory generation Yue et al. (2023); Huang et al.
 80 (2023); Yue et al. (2024), but they still need a reliable memory index when
 81 prior advice must be recovered from a growing same-domain log.

82 Legal consultation archives add a further difficulty: high lexical overlap
 83 among topically adjacent matters, with the professionally useful evidence
 84 unit being the full question–answer episode. This setting motivates *dual-*
 85 *store* memory architectures that keep fast episodic turn retrieval separate
 86 from slower semantic clustering, and session-aware soft binding operators
 87 that recover answer turns under same-domain interference without collapsing
 88 rank competition among unrelated candidates.

89 2.2. Dual-store memory and episodic retrieval

90 Dual-store designs distinguish fast episodic encoding from slower seman-
 91 tic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al.
 92 (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue
 93 agents extend context via hierarchical paging, summary banks, graphs, and
 94 neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen
 95 et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al.
 96 (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as
 97 flat passages rather than *multi-turn consultation episodes* whose answer turn
 98 may rank below an unrelated question turn sharing legal vocabulary. Long-
 99 MemEval and LoCoMo evaluate long-horizon recall and QA over personal
 100 conversation logs Wu et al. (2024); Maharana et al. (2024), while broader
 101 memory-agent surveys emphasize persistence beyond the context window Wu
 102 et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses
 103 reuse of historical search evidence across turns Mo et al. (2026). BIMS-
 104 LEGAL positions its dual-store contribution at this intersection: episodic
 105 turn embeddings with *sid* for same-session recovery (Soft O2), and BIRCH
 106 centroids with Soft O2-C for cross-session recovery when query and gold ev-
 107 idence occupy different sessions but share thematic structure Zhang et al.

(1996). Under dense same-domain interference, a question turn can occupy top- k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense-sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

Contrast with joint-episode indexing. An alternative way to keep episode context is to concatenate full question-answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance (βs) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the BIRCH pathway.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	Unit	Expansion	Type	Rank
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+ <i>sid</i>	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

143 In short, the dual-store split is the system design, and Soft O2 /Soft O2-C
 144 are the operators evaluated on Chinese legal consultation archives.

145 3. Method: BIMS-LEGAL

146 3.1. Architecture overview

147 Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complemen-
 148 tary episodic and semantic stores over the same consultation archive.

149 **Definition 1** (Episodic and semantic elements). *An episodic turn (episodic*
 150 *element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, sid_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a times-*
 151 *tamp, c_i conversational context (including speaker role), ϕ_i surface features,*
 152 *and sid_i the consultation session identifier. We reserve session turns for the*
 153 *weaker notion of turns that share only sid under a grouping key, without re-*
 154 *quiring role, timestamp, or other episodic metadata to participate in scoring.*
 155 *Soft O2 binds over session siblings $Sib(t) = \{t' : sid_{t'} = sid_t\} \setminus \{t\}$, but*
 156 *the indexed objects themselves remain episodic turns. A semantic element is*
 157 *$s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j , label κ_j , and formation*
 158 *statistics τ_j . Write $Clus(t) = \{t' : cid_{t'} = cid_t\} \setminus \{t\}$ for BIRCH-cluster*
 159 *siblings.*

160 **Definition 2** (Dual-store retrieval map). *Given query q with embedding*
 161 *$\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$*
 162 *under Eq. (1). Soft O2 produces $\mathcal{R}_{O2}(q) = \text{SoftBind}(\mathcal{R}_0(q); Sib, \beta)$ (Sec-*
 163 *tion 3.3). Soft O2-C produces $\mathcal{R}_{O2C}(q) = \text{SoftBind}(\mathcal{R}_0(q); Clus, \beta_c)$ (Sec-*
 164 *tion 3.5). Gated Hybrid composes session binding then cross-session cluster*
 165 *binding (Algorithm 2). The returned list is the top-k of the bound set under*
 166 *the rewritten scores.*

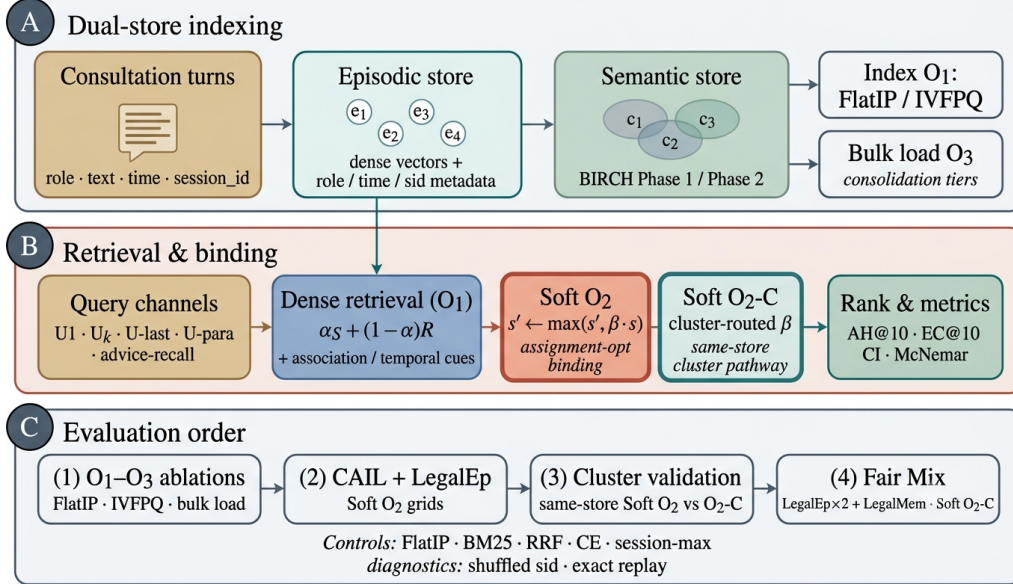


Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and BIRCH semantic clustering (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

3.2. Base scoring and implementation operators (O1, O3)

For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory turn m by

$$\text{score}(m) = \alpha S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha) R(t_m), \quad (1)$$

where S is cosine similarity on ℓ_2 -normalized embeddings and

$$R(t_m) = \max\left(0, 1 - \frac{\Delta_t}{30 \text{ days}}\right), \quad (2)$$

with Δ_t the elapsed time between query time and the turn timestamp. Unless stated otherwise, $\alpha=0.7$. Eq. (1) is the ranking score used throughout the legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step and do not replace Eq. (1).

O1: Exact FlatIP indexing.. Let $\hat{S}$ denote the similarity induced by an ANN backend. At hundreds to a few thousand turns, FAISS IVFPQ is often under-trained, so $\|\hat{S} - S\|$ is large enough to reorder turns and break Soft O2

178 inheritance. O1 forces $\hat{S} = S$ via exact FlatIP (inner-product search on
 179 ℓ_2 -normalized embeddings). Quantization error depresses Answer Hit (Ta-
 180 ble A.3: IVFPQ AH 0.330 at $M=1600$ on LegalEp-DISC exact). O1 is an
 181 engineering prerequisite for Soft O2 evaluation, not a standalone retrieval
 182 algorithm.

183 *O3: Bulk-load consolidation..* During bulk construction, O3 replaces the per-
 184 insert update ($e_i \mapsto \text{BIRCH}(e_i); \text{Flush}$) by a deferred operator $\text{Finalize} =$
 185 $\text{BIRCH}(\{e_i\}_{i=1}^N) \circ \text{Flush}$, executed once after all episodic writes. O3 does
 186 not change retrieval semantics; it makes $M \geq 400$ shared-store experiments
 187 runnable (≈ 11 min wall-clock for $M=400$; amortized ≈ 0.825 s/turn over
 188 ~ 800 turns). Quality–latency curves to $M=6400$ appear in Section 5.6.

189 3.3. Soft O2: session soft binding

190 *Motivation..* Under same-domain interference, a retrieved question turn can
 191 score highly while its answer sibling remains outside top- k —episode incom-
 192 pleteness compounded by IVFPQ score collapse and context fragmentation.
 193 Soft O2 completes episodes without hard score copy.

194 **Definition 3** (Soft binding operator). *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a*
 195 *sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\text{SoftBind}(\mathcal{R}; \mathcal{N}, \beta) : \quad s_{t'} \leftarrow \max(s_{t'}, \beta \cdot s_t) \quad \text{for all } t \in \mathcal{R}, t' \in \mathcal{N}(t). \quad (3)$$

196 *Soft O2 is $\text{SoftBind}(\cdot; \text{Sib}, \beta)$ with default $\beta=0.98$. Hard parent hydration is*
 197 *the special case $\beta=1$ with forced insertion of all siblings; session-max replaces*
 198 *each member score by $\max_{t \in \text{sid}} s_t$.*

199 Algorithm 1 states Soft O2 as sibling expansion followed by a complete-
 200 ness pass and top- k truncation, matching the public implementation.

201 3.4. Soft O2 attenuation and score competition

202 The attenuation β controls how strongly a retrieved turn promotes its
 203 session siblings. Availability of the injected sibling rises with β , while fidelity
 204 of the direct hit falls as $\beta \rightarrow 1$. Table 2 summarizes the empirical gap-ratio
 205 distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy:
 206 median ρ lies well below one, and a high- β operating point near 0.98 covers
 207 most measured distractors while remaining strictly below hard hydration
 208 ($\beta=1$). We therefore characterize the availability–ranking trade-off with a

Algorithm 1 Soft O2 session soft binding

Require: Query q , base hits $\mathcal{R}_0$, top- k , attenuation β

```
1:  $\mathcal{R} \leftarrow \mathcal{R}_0$ 
2: for each hit  $t \in \mathcal{R}_0$  with score  $s_t$  do
3:   for each sibling  $t' \in \text{Sib}(t)$  absent from  $\mathcal{R}$  do
4:     insert  $(t', \beta \cdot s_t)$  into  $\mathcal{R}$ 
5:   end for
6: end for
7:                                      $\triangleright$  Completeness pass after fusion with Eq. (1)
8: for each session  $sid$  represented in  $\mathcal{R}$  do
9:    $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R}, sid_u = sid\}$ 
10:  for each  $t' \in \text{Sib}$  of  $sid$  still absent do
11:    insert  $(t', \beta \cdot s_{\max})$  into  $\mathcal{R}$ 
12:  end for
13: end for
14: return top- $k$  of  $\mathcal{R}$  by descending score
```

Table 2: Gap-ratio statistics on LegalEp $M=3000$ stores (exact-channel query proxy; strongest measured non-session distractor). Default β is chosen by the AH sweep in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

209 three-candidate score-competition model and choose β by validation sweep
210 rather than by a closed-form optimum. A zero-margin hinge objective on
211 $\beta \in (0, 1]$ is uninformative for this choice: its rank term vanishes for all
212 $\beta < 1$.

213 **Definition 4** (Three-candidate score-competition model). *For a query whose*
214 *dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be*
215 *the gold answer sibling and let d be the strongest non-session distractor in a*
216 *wide candidate pool, with scores s_a and s_d . Soft O2 rewrites*

$$s'_h = s_h, \quad s'_a = \max(s_a, \beta s_h), \quad s'_d = s_d, \quad (4)$$

217 *and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured*
218 *distractor, conditions based on ρ are conservative sufficient conditions for*

219 *beating that distractor, not exact top-k boundary conditions for every com-*
 220 *petitor.*

221 The binding analysis is most relevant in the soft-injection regime $s_a \leq \beta s_h$,
 222 where $s'_a = \beta s_h$.

223 **Proposition 1** (Feasible attenuation band). *Under Definition 4, assume*
 224 *the soft-injection case $s_a \leq \beta s_h$. Then (i) availability against the measured*
 225 *distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the*
 226 *injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$,*
 227 *any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$*
 228 *already, rank preservation depends on the observed sibling score rather than*
 229 *on β alone.*

230 *Proof.* If $s_a \leq \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e.
 231 $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when
 232 $s_h > 0$. $\square$

233 *Soft-rank visualization and sweep.* On a validation grid we also plot the soft
 234 pairwise surrogate

$$\mathcal{L}_{\text{sr}}(\beta; \rho) = -\log \sigma(\tau(\beta - \rho)) - \gamma \log \sigma(\tau(1 - \beta)), \quad (5)$$

235 with logistic σ , temperature τ , and rank weight γ . The curve is a visualization
 236 of the availability–ranking trade-off under the measured ρ pool; it is not used
 237 as a fitted optimum. Default $\beta=0.98$ is taken from the AH sweep in Table A.4,
 238 where $\{0.90, 0.95, 0.98\}$ form a stable high-AH band on CAIL follow-up and
 239 LegalEp paraphrase as well as exact grids. Advice-recall uses the same default
 240 without a separate gap-ratio fit.

241 Fair controls include hard parent hydration, session-max aggregation, and
 242 shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max
 243 coincide under full-sibling expansion, so tables report a single hard-expansion
 244 baseline. Concatenating a full question–answer episode into one retrieval unit
 245 (BM25-joint or dense joint-episode documents) changes the evidence unit
 246 from turns to documents; we treat that design as an alternative indexing
 247 choice rather than a Soft O2 turn-binding baseline (Section 4).

248 3.5. Semantic consolidation and Soft O2-C

249 Semantic consolidation follows a BIRCH-style dual-phase procedure Zhang
 250 et al. (1996).

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c) , attenuations (β, β_c)

```
1:  $\mathcal{R} \leftarrow \text{SoftBind}(\mathcal{R}_0; \text{Sib}, \beta)$  ▷ session pathway first
2:  $\mathcal{S}_{\text{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$ 
3:  $\mathcal{T}_{\text{trig}} \leftarrow \{t \in \mathcal{R}_0\}$  ▷ direct dense hits only
4: for each trigger  $t \in \mathcal{T}_{\text{trig}}$  do
5:    $s_{\text{max}} \leftarrow \max\{s_u : u \in \mathcal{R}, cid_u = cid_t\}$ 
6:   for each  $t' \in \text{Clus}(t)$  with  $sid_{t'} \notin \mathcal{S}_{\text{cov}}$ , up to budget  $B_c$  do
7:      $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\text{max}})$ 
8:   end for
9: end for
10: return top- $k$  of  $\mathcal{R}$ 
```

251 **Definition 5** (BIRCH dual-phase consolidation). *Let $\{\mathbf{c}_j\}$ be current cen-*
252 *troids and θ_H, θ_M the assign /merge thresholds. **Phase 1 (assign).** For*
253 *a new embedding $\mathbf{v}$, set $j^* = \arg \max_j S(\mathbf{v}, \mathbf{c}_j)$; if $S(\mathbf{v}, \mathbf{c}_{j^*}) \geq \theta_H$ then*
254 *$C_{j^*} \leftarrow C_{j^*} \cup \{\mathbf{v}\}$ and refresh $\mathbf{c}_{j^*}$, else spawn a new cluster. **Phase 2***
255 ***(merge).** Periodically, merge pairs with $S(\mathbf{c}_i, \mathbf{c}_j) \geq \theta_M$ and purge near-*
256 *duplicates above a redundancy threshold. Under O3, both phases run inside*
257 **Finalize.** *Each turn t receives a cluster id cid_t , inducing $\text{Clus}(t)$ in Defini-*
258 *tion 1.*

259 **Definition 6** (Soft O2-C). *Soft O2-C is $\text{SoftBind}(\cdot; \text{Clus}, \beta_c)$ with default*
260 *$\beta_c=0.95 < \beta$, reflecting noisier thematic neighbourhoods than session mem-*
261 *bership. A per-cluster sibling budget B_c caps $|\text{Clus}(t) \cap \text{Injected}|$ so large*
262 *thematic clusters cannot flood top- k .*

263 Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings
264 relative to sessions already covered by dense /Soft O2 hits, and triggers
265 cluster binding only on direct dense hits. This prevents thematic confusers
266 from displacing same-session siblings already recovered by Soft O2. Soft O2
267 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway
268 operator when gold evidence lies in another session of the same thematic
269 cluster.

270 4. Corpora and Evaluation Protocol

271 Figure 2 sketches the shared-corpus design used throughout the legal eval-
272 uation. Gold needles and same-domain distractors occupy one store. Each

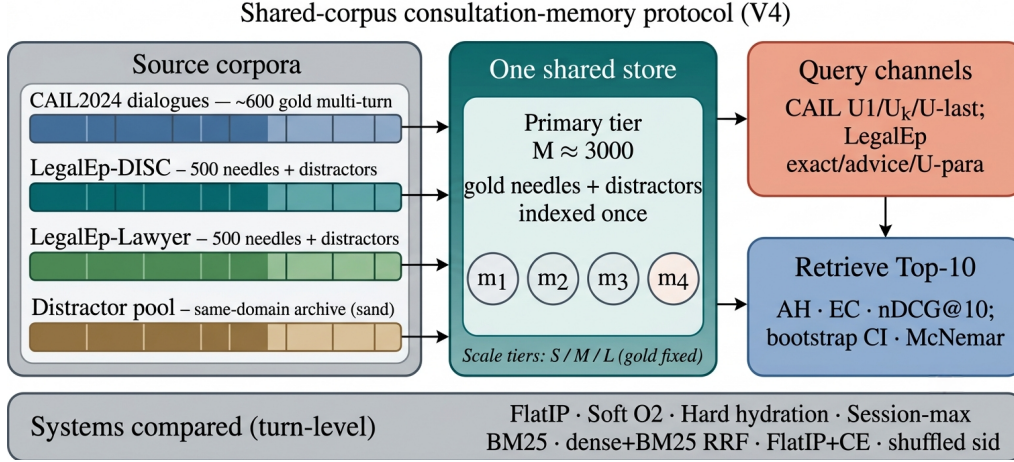


Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

query probes that shared archive under fixed interference conditions. Privately rebuilt haystacks are avoided so that system comparisons isolate the retrieval operator.

4.1. Datasets and query channels

Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues. Gold needles are authentic preliminary and final dialogues, with conversation turns completed by label turns where applicable. Query channels cover the first user turn ($U1$), a later in-dialogue user turn ($U_k\text{-followup}$), the last user turn ($U\text{-last}$), and constrained paraphrases ($U\text{-para}$). Evidence defaults to assistant turns in the target session.

LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is a natural (question, answer) pair with session identifier *sid*. Reconstruction applies length bounds and legal-content filters, excludes exam prompts, removes near-duplicates, separates needles from distractors under a fixed seed, and retains an audit subsample. After filtering, each LegalEp corpus retains 3500 passed sessions (500 needles, 2500 distractors at the $M=3000$ primary tier).

Query-channel policy. Exact replay uses the stored user question and is reported only as a *diagnostic* upper bound on surface-form recovery. Primary

claims use paraphrase, Uk-followup /U-last, and advice-recall queries whose surface form is not identical to the stored question text; paraphrase and advice-recall templates are generated under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use $M=400$, $S=300$.

4.2. Metrics and baselines

Primary metrics are Answer Hit@ k (AH), Episode Completeness@ k (EC), and normalized Discounted Cumulative Gain@ k (nDCG) at $k=10$. AH is binary: whether a gold answer turn appears in the top- k . EC is the mean coverage of gold turns within the target session. Session Hit@ k is reported only for diagnosis. For nDCG we use fixed graded relevance over the full gold session: answer turns $rel=1.0$, other same-session turns $rel=0.5$, all other turns $rel=0.0$. The ideal denominator (IDCG) is computed from all gold-relevant turns in the target session and truncated at k , so every system on a query shares the same IDCG. Answer Hit and Episode Completeness are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session IDCG so that systems remain comparable on each query. Hard expansion can raise nDCG when copied siblings occupy early ranks, so AH and EC are read together with nDCG.

The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL and LegalEp transfer channels; for that family we report bootstrap 95% CIs, McNemar mid- p , and Holm-adjusted p -values (Table A.2). Hard-hydration, CE, and Soft O2-C /Hybrid Mix contrasts are secondary.

Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies soft sibling score inheritance on the episodic pathway; Soft O2-C applies the same rule within BIRCH clusters. Hard parent hydration expands a hit to all siblings by copying the trigger score. Lexical controls include BM25 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve pre-concatenated episode documents and therefore answer a coarser retrieval problem; they are discussed as alternative designs in Section 6.3 rather than Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the top-30 FlatIP candidates with BAAI/bge-reranker-v2-m3. Shuffled *sid*,

329 a β sweep, and IVFPQ provide membership, attenuation, and quantization
 330 controls.

331 4.3. Fair Mix protocol for Soft O2-C

332 On fully same-session LegalMem grids Soft O2 already recovers gold an-
 333 swers that share *sid*, leaving Soft O2-C little exclusive headroom and expos-
 334 ing it to thematic confusers. To study off-session recovery we rebuild the *same*
 335 LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes
 336 are split into query-side S_a and evidence-side S_b with distinct session identi-
 337 fiers, while 30% remain intact. Distractors are unchanged. FlatIP, Soft O2,
 338 Soft O2-C, and gated Hybrid share unrestricted dense retrieval on this store;
 339 we report overall AH and stratum AH on `cross_session` vs `same_session`
 340 queries.

341 *Cluster co-membership and purity under Mix.* After BIRCH consolidation,
 342 each cross-session Sa/Sb pair is assigned a shared cluster identifier. This
 343 construction uses split metadata available only when building the evaluation
 344 store; it does not inject labels at query time. The intent is to test Soft O2-
 345 C when thematic co-membership is present. Cluster purity still matters: a
 346 shared *cid* that also contains many thematically adjacent distractors narrows
 347 Soft O2-C’s effective margin, which is why Hybrid gates cluster injection to
 348 direct dense hits and caps the per-cluster budget B_c . If recovery still fails
 349 under guaranteed Sa/Sb co-membership, the failure is attributable to binding
 350 rather than accidental cluster separation. Natural same-cluster rates without
 351 this construction are lower (Section 6.3). Soft O2’s session membership rule
 352 is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses
 353 mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior
 354 runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon
 355 memory context only Wu et al. (2024); Maharana et al. (2024).

356 5. Experiments

357 Unless noted, all dense retrieval grids use the multilingual encoder `nomic-embed-text-v2-moe`
 358 Nussbaum et al. (2025) (768-d) as a single shared embedding backend, with
 359 $k=10$ and seed 42 on the main $M \approx 3000$ stores. We keep the encoder fixed
 360 so Soft O2 vs FlatIP contrasts isolate binding rather than representation
 361 changes; sensitivity to alternative Chinese or multilingual encoders (e.g.,
 362 `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed check on a

363 smaller shared store ($M=400$) is reported later. Primary contrasts appear in
 364 Figures 3–5; full numeric grids are in the appendix. Each main table maps
 365 to a released JSON artifact in the accompanying repository.

366 5.1. Implementation operators and Soft O2 ablation

367 Table 3 reports the shared-store ablation on DISC-Law and Lawyer-LLaMA
 368 ($M=400$, $S=300$). IVFPQ baseline Answer Hit is 0.540 / 0.397. O1 alone
 369 yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397), confirm-
 370 ing that exact indexing is a necessary implementation prerequisite but insuffi-
 371 cient alone. O1+Soft O2 raises Answer Hit to 0.780 / 0.680 (+0.240 / +0.283
 372 over baseline) and Episode Completeness to 0.888 / 0.840. O3 is enabled for
 373 all configurations in this ladder as a construction prerequisite; it does not
 374 alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store ($M=400$, $S=300$, $k=10$). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M \approx 3000$ grids appears in Table 6.

Corpus	System	EC@10	AH@10
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	0.888	0.780
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	0.840	0.680

375 5.2. Failure taxonomy

376 We classify each query into *complete* (answer evidence in top- k), *incom-*
 377 *plete* (target session partially retrieved but answer missing), or *session miss*
 378 (no target-session turn in top- k). Table 4 contrasts FlatIP, Soft O2, hard
 379 hydration, and shuffled *sid* on primary non-exact channels. Soft O2 converts
 380 incompleteness into complete retrieval while session-miss stays nearly fixed;
 381 shuffled *sid* restores high incompleteness, so the gain is session-tied rather
 382 than generic score inflation. Hard hydration can also drive incompleteness
 383 near zero by copying trigger scores, but it is not Soft O2’s binding rule and
 384 can over-promote siblings (Section 5.4).

Table 4: Failure taxonomy ($k=10$). Boldface: best complete / lowest incomplete within each corpus-channel block among FlatIP, Soft O2, and shuffled *sid* (hard hydration shown as a score-copy envelope). Soft O2 cuts incompleteness without raising session miss; shuffled *sid* undoes the gain.

Corpus / channel	System	Complete	Incomplete	Session miss
DISC / u-param	FlatIP	70.8%	26.8%	1.6%
	Soft O2	89.1%	8.7%	1.8%
	Hard hydr.	97.2%	0.0%	2.8%
	Shuffled O2	66.4%	31.4%	2.2%
DISC / advice	FlatIP	48.0%	17.0%	35.0%
	Soft O2	60.8%	3.8%	35.4%
	Hard hydr.	65.0%	0.0%	35.0%
	Shuffled O2	45.6%	19.2%	35.2%
Lawyer / u-param	FlatIP	71.5%	25.3%	0.8%
	Soft O2	94.0%	3.8%	1.6%
	Hard hydr.	98.0%	0.0%	2.0%
	Shuffled O2	72.9%	25.5%	1.6%
CAIL / Uk	FlatIP	45.8%	54.0%	0.2%
	Soft O2	92.8%	7.0%	0.2%
	Hard hydr.	98.3%	1.5%	0.2%
	Shuffled O2	34.7%	65.2%	0.2%

385 5.3. Turn-level Soft O2 controls

386 Primary Soft O2 claims use a shared turn-level index. Table 5 reports An-
 387 swer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary
 388 Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel;
 389 shuffled *sid* collapses toward or below FlatIP; hard hydration is competitive
 390 on CAIL AH but Soft O2 remains stronger on LegalEp paraphrase /advice
 391 and retains higher EC in the full grids (Tables A.5–A.7). BM25-turn is a
 392 same-granularity lexical control (Table A.8): Soft O2 exceeds it on CAIL
 393 Uk /U-last (0.698/0.762 vs 0.482/0.413). Joint episode / document-level in-
 394 dexes change evidence granularity and are not Soft O2 protocol competitors.

395 5.4. Soft O2 on session (CAIL and LegalEp)

396 Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn
 397 channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over FlatIP
 398 on every channel (U1 0.788 vs 0.570**; Uk 0.698 vs 0.270**; U-last 0.762 vs

Table 5: Primary Soft O2 controls (AH@10, $M \approx 3000$). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is the session-binding claim; shuffled *sid* tests membership dependence; hard hydration is score-copy expansion.

Corpus / channel	FlatIP	Soft O2	Hard	Shuffled
CAIL / U1	0.570	0.788	0.802	0.452
CAIL / Uk	0.270	0.698	0.720	0.230
CAIL / U-last	0.245	0.762	0.790	0.200
LegalEp-DISC / u-para	0.547	0.648	0.592	0.469
LegalEp-DISC / advice	0.342	0.428	0.402	0.310
LegalEp-Lawyer / u-para	0.568	0.759	0.596	0.592
LegalEp-Lawyer / advice	0.332	0.472	0.394	0.348

CAIL2024 multi-turn results ($M \approx 3000$, turn-level systems)

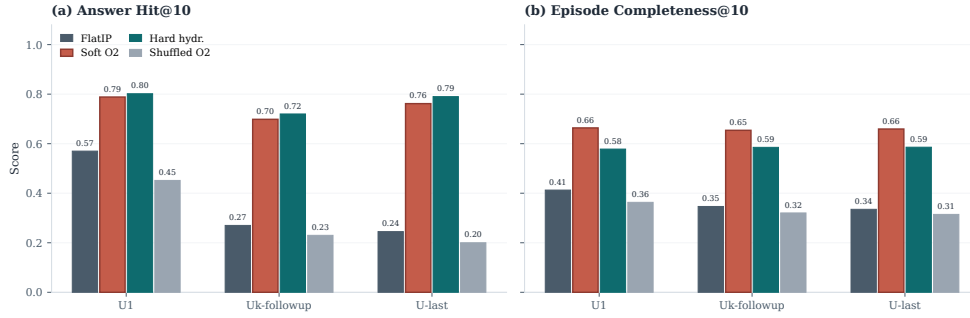


Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can maximize AH at a modest EC cost.

399 0.245**). Shuffled *sid* removes the gain (U1 0.452; Uk 0.230; U-last 0.200),
400 tying the improvement to correct episode membership. Hard expansion can
401 raise AH slightly above Soft O2 (e.g., U1 0.802 vs 0.788) while lowering EC
402 (U1 0.578 vs Soft O2 0.664). Soft O2 therefore behaves as a rank-preserving
403 binder: siblings enter under attenuated scores, and episode coverage remains
404 higher than under hard score copy. Table 6 shows the Soft O2>FlatIP order-
405 ing under graded nDCG with gold-session IDCg, while incompleteness falls
406 sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

407 Figure 4 reports LegalEp transfer on the same AH grids. On LegalEp-
408 DISC exact replay Soft O2 and FlatIP stay near-tied (0.638 vs 0.640), where
409 both turns are already recoverable for many queries—consistent with treat-
410 ing exact as diagnostic. On Lawyer exact Soft O2 improves AH to 0.752

Table 6: Graded nDCG@10 with gold-session IDCG (answer $rel=1.0$, other same-session turns $rel=0.5$) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	0.793	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	0.791	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	0.759	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	0.524	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	0.844	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	0.582	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

411 from FlatIP 0.692 (McNemar mid- $p=0.024^*$). Paraphrase yields the clearest
412 LegalEp gains: Soft O2 reaches 0.648 vs FlatIP 0.547 on DISC and 0.759 vs
413 0.568 on Lawyer, exceeding hard expansion on both corpora (mid- $p<0.001^{**}$
414 on both corpora). Advice-recall queries omit the stored question surface form:
415 Soft O2 raises AH to 0.428 from FlatIP 0.342 on DISC and to 0.472 from
416 0.332 on Lawyer, again above hard expansion (mid- $p<0.001^{**}$), while shuffled
417 *sid* falls toward or below FlatIP. Graded nDCG mirrors the Soft O2>FlatIP
418 ordering on paraphrase and advice-recall (Table 6). Detailed grids are in
419 Tables A.6–A.7.

420 Cross-encoder reranking is a strong neural control at the same turn-
421 level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with
422 **bge-reranker-v2-m3**. Table 7 shows that FlatIP+CE is competitive—and
423 on LegalEp-DISC exact, strongest—when surface-form overlap already places
424 both question and answer near the top of the dense pool (DISC exact AH
425 0.722 for FlatIP+CE vs 0.638 for Soft O2). On channels with semantic shift
426 or multi-turn drift, Soft O2 pulls ahead: CAIL Uk 0.698 vs FlatIP+CE 0.375,

LegalEp ($M = 3000$): Soft O2 gains are strongest on paraphrase and remain positive on advice-recall

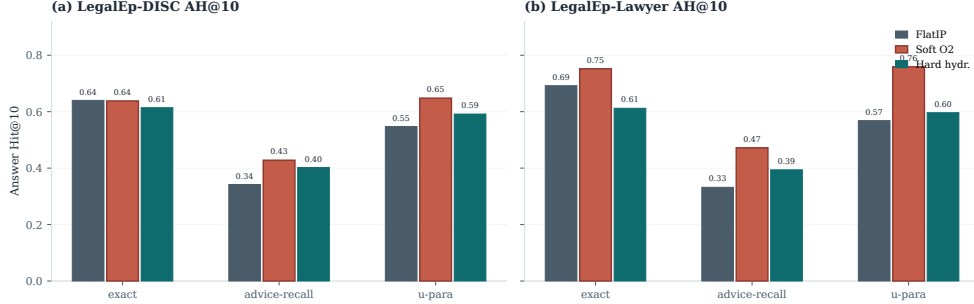


Figure 4: LegalEp AH@10 ($M=3000$, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Cross-encoder control (AH@10). FlatIP+CE reranks the top-30 FlatIP candidates with **bge-reranker-v2-m3**. Boldface: best AH within each row.

Corpus / channel	FlatIP	FlatIP+CE	Soft O2
CAIL / U1	0.585	0.688	0.788
CAIL / Uk	0.265	0.375	0.698
CAIL / U-last	0.228	0.342	0.762
LegalEp-DISC / exact	0.572	0.722	0.638
LegalEp-Lawyer / exact	0.636	0.708	0.752

427 CAIL U-last 0.762 vs 0.342, and LegalEp-Lawyer exact 0.752 vs 0.708. A
 428 structural limit of the reranker explains the split: CE can refine ranking
 429 among candidates already retrieved by FlatIP, but it cannot promote an
 430 answer sibling that never entered the top-30 pool. Soft O2 recovers such
 431 siblings by propagating attenuated scores through session links, which helps
 432 precisely when the stored advice turn has weak direct lexical overlap with
 433 the query (paraphrase and advice-recall; cf. Tables 5 and 6).

434 5.5. Soft O2-C and gated Hybrid under Mix

435 When gold answers already share *sid* with the query, Soft O2 is sufficient
 436 and cluster binding is unnecessary. Table 8 asks the off-session question
 437 under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and
 438 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated
 439 Hybrid sharing unrestricted dense retrieval.

On the cross-session stratum Soft O2-C improves over Soft O2 on LegalEp-Lawyer (0.491 vs 0.446) and LegalMem-MT (0.537 vs 0.503): cluster binding can surface split-episode gold that session membership alone misses. Gated Hybrid converts that mechanism into an overall gain on LegalEp-Lawyer (0.534 vs Soft O2 0.496; mid- $p=0.010^*$) and a directional gain on LegalMem-MT (0.646 vs 0.618), while preserving most same-session hits (0.560 / 0.867). On LegalEp-DISC the Hybrid edge is negligible (0.486 vs 0.482). Ungated Soft O2-C can raise cross-session hits yet displace intact same-session evidence, which is why Hybrid triggers cluster binding only on direct dense hits. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Appendix Appendix A, Table A.1).

Table 8: Mix evaluation of Soft O2-C and gated Hybrid (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	System	AH	AH _{cross}	AH _{same}	mid- p vs Soft O2
LegalEp-DISC Mix	FlatIP	0.478	0.477	0.480	—
	Soft O2	0.482	0.443	0.573	—
	Soft O2-C	0.410	0.443	0.333	0.0004**
	Hybrid	0.486	0.469	0.527	0.8099
LegalEp-Lawyer Mix	FlatIP	0.462	0.454	0.480	—
	Soft O2	0.496	0.446	0.613	—
	Soft O2-C	0.464	0.491	0.400	0.0857
	Hybrid	0.534*	0.523	0.560	0.0105*
LegalMem-MT Mix	FlatIP	0.534	0.526	0.553	—
	Soft O2	0.618	0.503	0.887	—
	Soft O2-C	0.500	0.537	0.413	0.0000**
	Hybrid	0.646	0.551	0.867	0.1837

McNemar mid- p on paired Answer Hit; * $p<0.05$, ** $p<0.01$. Cluster co-membership of Sa/Sb pairs follows the Mix protocol in Section 4.3.

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on LegalEp-DISC exact ($Q=100$, three repeats; $M \in \{100, 400, 1600, 6400\}$). Soft O2 retains an AH advantage at each tier; IVFPQ collapses at $M=1600$ (AH 0.330), motivating O1; absolute latency grows with M . We treat O1 FlatIP

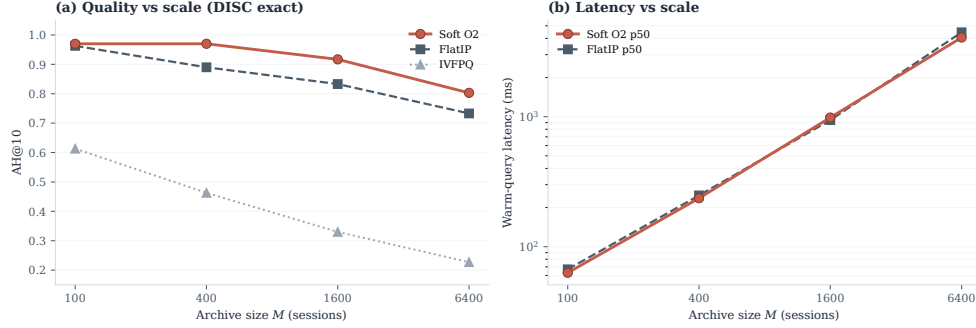


Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M .

as an engineering prerequisite for Soft O2 at these scales and avoid claiming deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit ($N=270$ per corpus; generator **qwen3:14b**, independent judge **qwen3:32b**) reports judged answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs Soft O2 generation experiment. Full protocol and stratified results appear in Section 6.2. On a smaller shared store ($M=400$, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040 .

6. Discussion

6.1. Findings and scope

Results show that episode incompleteness is a substantial FlatIP failure mode under same-domain interference, and that Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio analysis (Section 3.4) motivates keeping β close to but below unity; the AH sweep peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration ($\beta=1$). On CAIL, Soft O2 raises Answer Hit by large margins (U1 +0.218, Uk +0.428, U-last +0.517) with collapsing shuffled-*sid* controls, and keeps higher EC than hard expansion despite occasional AH ties. After Holm adjustment on the Soft O2 vs FlatIP family, the non-exact CAIL and LegalEp gains remain significant, while LegalEp exact replay is mixed (Table A.2). LegalEp

paraphrase and advice-recall, rather than exact replay, support the main transfer result. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). Under Mix, gated Hybrid recovers additional cross-session gold on LegalEp-Lawyer with a directional lift on LegalMem-MT (Table 8); Soft O2 remains preferable when gold already shares *sid* (Table A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes are a coarser design choice outside this protocol.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator **qwen3:14b** and independent judge **qwen3:32b** are distinct models. Per corpus we pool $N=270$ judged instances as P1 $n=120$ plus P2 $n=150$ (folder **legal_qa_n270**). The independent judge assigns a continuous evidence-answerability score in $[0, 1]$ from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score ≥ 0.7 . Table 9 reports pooled EAR and the mean judge score on a stratified subsample ($n=90$ /corpus; 30 each of exact, paraphrase, and follow-up), labelled “Stratified judge” in the table. In the complete-episode stratum—both question and answer turns in top- k —judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit ($n=60$ /corpus) separates retrieval failures (gold episode absent from top- k) from generation failures given retrieved evidence. Annotators used a binary answerability label on each instance; agreement is Cohen’s κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client’s matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure ev-

516 idence availability and judged answerability respectively; statutory correct-
 517 ness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit ($N=270$ per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7 . Stratified judge: mean judge score on $n=90$ /corpus (exact/paraphrase/follow-up). Human κ : Cohen’s κ on binary answerability labels from two blind annotators ($n=60$ /corpus).

Corpus	Pooled EAR	Stratified judge	Human κ
DISC-Law	0.867	0.757	0.71
Lawyer-LLaMA	0.859	0.729	0.68

518 6.3. Limitations

519 LegalEp episodes are consultation pairs; multi-turn authenticity remains
 520 reserved for CAIL.

521 *Dependency on session segmentation and cluster quality..* Soft O2 relies on
 522 correct session identifiers (*sid*). In uncured client logs, boundary noise—
 523 mid-session topic shifts, multi-intent interruptions, or automated segmenta-
 524 tion drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated
 525 inheritance may propagate false context; if a true advice turn is split into
 526 another session, Soft O2 cannot bind it. Soft O2-C likewise depends on
 527 BIRCH cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared
 528 cluster id so binding can be tested when thematic co-membership is present
 529 (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink
 530 the margin over dense retrieval. Deployments on raw consultation streams
 531 therefore need reliable upstream session segmentation, and Soft O2-C should
 532 be treated as gated rather than always-on.

533 *Joint-episode indexing..* Concatenating full question–answer pairs into
 534 episode documents prevents incompleteness by changing the evidence unit
 535 from turns to merged documents (Section 2). That design is outside the
 536 Soft O2 turn-binding protocol; a matched comparison under online updates,
 537 latency, and fine-grained advice isolation is left to future work.

538 *Embedding model and other scope limits..* All main grids use
 539 `nomic-embed-text-v2-moe`. A Chinese-specialized encoder such as
 540 `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ

distribution, and might shrink or widen Soft O2’s relative gain wherever the bi-encoder already places answer turns inside top- k . We did not re-run the full Soft O2 grids under alternative encoders; the binding operator itself is encoder-agnostic once fused scores are available. The ρ statistics use an exact-channel query proxy and the strongest measured distractor; paraphrase and advice-recall gap laws may shift the feasible band slightly, though the AH sweep keeps a stable high- β region on those channels. Main $M \approx 3000$ tables use a fixed seed; five-seed variance is reported only for the smaller $M=400$ store. LLM paraphrases need human audit. Statutory correctness, citation validity, and professional liability remain unevaluated. Because many channels are tested, Holm-adjusted p -values accompany the Soft O2 vs FlatIP family (Table A.2); Hybrid Mix gains outside LegalEp-Lawyer are treated as directional.

Latency and deployment scope.. Scale curves cover $M \in \{100, 400, 1600, 6400\}$ under a fixed hardware stack (Table A.3). At $M=6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL should therefore be positioned as **high-precision offline or quasi-real-time consultation memory recovery**—e.g., pre-meeting case review, audit replay, or batch RAG over a firm’s advice archive—not as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly 10^5 sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

7. Conclusion

BIMS-LEGAL recovers prior advice from Chinese legal consultation archives under same-domain interference with a dual store: an episodic turn index and a BIRCH semantic store. Soft O2 is the main algorithmic contribution: attenuated session binding completes episodes without hard score copy. Soft O2-C and gated Hybrid extend the same idea to thematic clusters when Mix reconstructions place some gold off-session. Attenuation β is chosen by validation sweep under a three-candidate score-competition account; O1 FlatIP and O3 bulk-load are the implementation operators that

make the shared-store ablations feasible. On CAIL multi-turn and LegalEp paraphrase /advice-recall channels, Soft O2 improves Answer Hit over FlatIP, keeps higher episode completeness than hard expansion on CAIL, and raises graded nDCG while cutting incompleteness; shuffled-*sid* controls collapse the gains. Under Mix, gated Hybrid recovers additional cross-session gold on LegalEp-Lawyer while Soft O2 remains preferable on intact same-session gold. An end-to-end Soft O2 QA audit associates complete episode retrieval with judged answerability and low wrong-session contamination, without claiming paired generation superiority over FlatIP. Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives; joint-episode indexing remains a coarser design outside this protocol.

Declarations

Funding

Computational resources were provided by the Data Law Laboratory, China University of Political Science and Law.

Conflict of interest

The authors declare no competing interests.

Ethics approval

This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory retrieval.

Data availability

Code, processed evaluation corpora, and primary result JSON files are available at <https://github.com/Kilimajaro/soft-episode-binding-legal-memory> (archived release to accompany camera-ready). Upstream corpora remain under their original licenses: CAIL2024 consultation tracks; Hugging Face ShengbinYue/DISC-Law-SFT and Skepsun/lawyer_llama_data. Processed LegalEp /LegalMem artifacts are derived research releases; redistributors should consult each upstream licence before reuse.

Code availability

The BIMS-LEGAL implementation (O1 FlatIP, Soft O2, O3 bulk-load, Soft O2-C), corpus builders, Mix constructors, and table/figure scripts ship in the same repository, with one JSON artifact per main table.

609 *Author contributions*

610 L.X. (University of Winnipeg) designed BIMS-LEGAL and the
611 Soft O2 /Soft O2-C evaluation, implemented the system, ran the experi-
612 ments, and wrote the manuscript. L.H. (corresponding author; CUPL Data
613 Law Lab and Institute for Data Law) provided methodology guidance, revi-
614 sion, and supervision.

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733 **Appendix A. Numeric grids and controls**

734 Same-session Soft O2-C contrasts, significance tests, scale curves, and full
 735 Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	Soft O2	Soft O2-C
LegalEp-DISC / exact	0.836	0.744
LegalEp-DISC / u-para	0.807	0.730
LegalEp-DISC / advice	0.548	0.490
LegalEp-Lawyer / exact	0.818	0.710
LegalEp-Lawyer / u-para	0.817	0.733
LegalEp-Lawyer / advice	0.546	0.448
LegalMem-MT / U1	0.903	0.712
LegalMem-MT / Uk	0.793	0.385

Table A.2: McNemar mid- p on paired Answer Hit (Soft O2 vs FlatIP / hard hydration / FlatIP+CE). Holm-adjusted p is reported for the nine primary Soft O2 vs FlatIP contrasts.

Corpus / channel	Contrast	ΔAH	mid- p	Holm p
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	8.09e-47
CAIL / Uk	O2 vs Hard hydr.	-0.022	0.243	—
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	4.67e-17
CAIL / U1	O2 vs Hard hydr.	-0.013	0.435	—
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
CAIL / U-last	O2 vs Hard hydr.	-0.028	0.0982	—
CAIL / Uk	O2 vs FlatIP+CE	+0.323	2.50e-26	—
CAIL / U1	O2 vs FlatIP+CE	+0.100	3.51e-05	—
CAIL / U-last	O2 vs FlatIP+CE	+0.420	3.92e-41	—
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.934	0.934
LegalEp-DISC / exact	O2 vs Hard hydr.	+0.024	0.302	—
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.0239	0.191
LegalEp-Lawyer / exact	O2 vs Hard hydr.	+0.140	3.50e-10	—
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	6.36e-04
LegalEp-DISC / u-para	O2 vs Hard hydr.	+0.056	0.035	—
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.68e-14
LegalEp-Lawyer / u-para	O2 vs Hard hydr.	+0.163	7.57e-09	—
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	8.75e-04
LegalEp-DISC / advice	O2 vs Hard hydr.	+0.026	0.215	—
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	9.44e-10
LegalEp-Lawyer / advice	O2 vs Hard hydr.	+0.078	5.72e-04	—

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	Mode	AH@10	Build (s)	p50 (ms)	p95 (ms)
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	25.2	40	54
100	Soft O2	0.970	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	171	204
400	Soft O2	0.970	30.9	236	313
1600	FlatIP	0.833	359.8	941	1278
1600	IVFPQ	0.330	251.4	669	751
1600	Soft O2	0.917	393.7	985	1172
6400	FlatIP	0.733	4722.8	4458	4969
6400	IVFPQ	0.227	1072.1	2182	3021
6400	Soft O2	0.803	5245.7	4061	4503

Table A.4: Soft O2 AH@10 versus β on the primary shared-corpus evaluation. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation ($\beta=0.5$) weakens binding.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid- $p < 0.01$. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10	95% CI (AH)
U1	FlatIP	0.570	0.413	0.554	[0.530,0.610]
	Soft O2	0.788	0.664	0.776	[0.757,0.820]
	Hard hydr.	0.802	0.578	0.864	[0.768,0.835]
	Shuffled O2	0.452	0.363	0.411	[0.410,0.492]
Uk-followup	FlatIP	0.270	0.347	0.445	[0.237,0.307]
	Soft O2	0.698	0.654	0.793	[0.660,0.735]
	Hard hydr.	0.720	0.585	0.847	[0.685,0.753]
	Shuffled O2	0.230	0.320	0.369	[0.198,0.262]
U-last	FlatIP	0.245	0.335	0.420	[0.212,0.280]
	Soft O2	0.762	0.659	0.791	[0.725,0.797]
	Hard hydr.	0.790	0.586	0.854	[0.757,0.822]
	Shuffled O2	0.200	0.315	0.348	[0.172,0.232]

736

Soft O2 leads EC; hard hydration can lead AH.

Table A.6: LegalEp-DISC ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p < 0.05$, ** $p < 0.01$ (Soft O2 vs FlatIP, McNemar mid- p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCg.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.640	0.741	0.730
	Soft O2	0.638	0.780	0.813
	Hard hydr.	0.614	0.598	0.886
	Shuffled O2	0.448	0.645	0.666
advice-recall	FlatIP	0.342	0.438	0.466
	Soft O2	0.428	0.503	0.524
	Hard hydr.	0.402	0.405	0.572
	Shuffled O2	0.310	0.421	0.436
u-param	FlatIP	0.547	0.681	0.681
	Soft O2	0.648	0.735	0.759
	Hard hydr.	0.592	0.571	0.827
	Shuffled O2	0.469	0.610	0.627

Table A.7: LegalEp-Lawyer ($M=3000$, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. * $p<0.05$, ** $p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.692	0.806	0.753
	Soft O2	0.752	0.848	0.869
	Hard hydr.	0.612	0.615	0.897
	Shuffled O2	0.570	0.747	0.705
advice-recall	FlatIP	0.332	0.464	0.483
	Soft O2	0.472	0.540	0.582
	Hard hydr.	0.394	0.403	0.601
	Shuffled O2	0.348	0.467	0.454
u-param	FlatIP	0.568	0.718	0.744
	Soft O2	0.759	0.819	0.844
	Hard hydr.	0.596	0.594	0.864
	Shuffled O2	0.592	0.712	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	Channel	AH@10
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460